

Making China AI Compliance Evidence Machine-Readable: A Compliance-as-Code Framework for Generative AI and Deep Synthesis Services

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Abstract—AI governance increasingly depends on evidence that can be reviewed, traced, and reused across legal, operational, and engineering workflows. In China-facing AIGC and deep synthesis services, relevant obligations may involve model filing or disclosure, generated-content labeling, consent and scope management, content safety review, complaint handling, and personal information protection. In practice, however, these materials are often maintained as static documents, screenshots, spreadsheets, or disconnected operational logs. This paper presents the China AIGC Compliance Evidence Generator, a compliance-as-code research prototype that converts selected governance obligations into machine-readable controls and request-level evidence artifacts. The prototype models a synthetic AIGC actor and digital human advertising platform and evaluates each generation request against controls for consent validity, commercial-use scope, model filing metadata, visible and metadata-based labeling, content safety, and complaint handling. A seeded validation scenario assessed 1,000 synthetic generation requests, producing 800 pass, 120 review, and 80 fail outcomes; the system also generated risk records, remediation items, and 88 selected per-request evidence bundles. The prototype does not certify legal compliance and uses no real personal, biometric, face, voice, contract, or company data. Its contribution is a feasibility demonstration: selected China-facing AIGC governance concerns can be operationalized as machine-readable evidence pipelines rather than reconstructed after the fact through fragmented documentation.

Index Terms—AI governance, AI compliance, compliance-as-code, AIGC, deep synthesis, synthetic data, evidence engineering, privacy engineering.

I. INTRODUCTION

AI governance frameworks describe what organizations should assure, but they often leave open how evidence should be generated, structured, and reviewed. The result is an implementation gap between governance requirements and operational systems. Machine-readable compliance evidence addresses this gap by representing obligations, controls, assessments, and supporting artifacts in formats that can be validated and reused by software [1].

This paper adapts that logic to a China-facing AIGC service context. Chinese AI governance emphasizes generative AI service governance, deep synthesis management, AIGC labeling, algorithmic recommendation governance, model filing or registration disclosure, platform responsibility, complaint

handling, and personal information protection [2], [3], [4], [5], [6]. These obligations are not identical to high-risk conformity assessment under the EU AI Act. They are often service- and platform-centered, and they require evidence about content labeling, operational safeguards, complaint handling, and use authorization.

The technical prototype described here models an AIGC actor and digital human advertising platform. Clients submit generation requests using licensed virtual actor assets. The system evaluates whether each request has valid actor consent, whether the requested commercial use is within consent scope, whether the relevant model has filing metadata, whether the output has visible and invisible labels, whether content safety checks are appropriate, and whether content complaints are handled within a synthetic service-level threshold.

The contribution is a concrete evidence pipeline rather than a legal conclusion. The prototype uses synthetic data only. It does not use real personal data, biometric data, face data, voice data, contracts, company records, or production media. Its purpose is to demonstrate the feasibility of converting selected obligations into machine-readable controls, assessment results, risk records, remediation items, and per-request evidence bundles.

A. Related Work: From Documentation to Evidence Pipelines

Prior work on AI documentation established important predecessors for structured transparency. Model Cards provide a template for model reporting, including intended use, performance characteristics, limitations, and ethical considerations [7]. FactSheets support supplier declarations and AI service transparency by encouraging providers to document purpose, performance, safety, security, and provenance [8]. Datasheets for Datasets improve dataset documentation by recording motivation, composition, collection, preprocessing, distribution, maintenance, and provenance [9].

These formats are useful for transparency and review, but they are mostly static artifacts. They do not by themselves create continuous, request-level operational evidence tied to individual service events, labeling checks, consent status, complaint handling, and remediation workflows. A model card can describe a model, and a datasheet can describe a dataset, but an AIGC service may also need evidence that a specific generation request used an authorized actor asset,

passed content checks, produced labeled output, and triggered an appropriate complaint or remediation path.

Risk-management and management-system frameworks provide another foundation. The NIST AI Risk Management Framework gives organizations vocabulary and functions for governing, mapping, measuring, and managing AI risk [10]. AI governance maturity models build on this idea by evaluating how well organizations operationalize risk-management practices [11]. ISO/IEC 42001 provides an AI management system structure for organizational governance [12]. These frameworks are valuable for governance design, but neither provides a concrete request-level evidence pipeline for a China-facing AIGC service.

OSCAL is relevant because it shows how controls, assessment plans, assessment results, and remediation artifacts can become machine-readable in cybersecurity compliance [13]. The source paper on machine-readable AI compliance evidence extends this logic to EU AI Act high-risk systems, arguing that compliance evidence should be represented as structured artifacts rather than reconstructed from prose after the fact [1].

This paper follows that direction but adapts the evidence-pipeline logic to a different regulatory and operational setting: China-facing AIGC actor and digital human services. In this setting, selected obligations may require evidence for model filing or disclosure, visible and metadata-based labeling, consent scope, content safety, complaints, and remediation. The prototype therefore focuses on request-level evidence artifacts rather than general-purpose AI documentation alone.

II. REGULATORY BACKGROUND: CHINESE AI GOVERNANCE AS EVIDENCE OBLIGATIONS

Chinese AI governance can be interpreted as a set of evidence obligations for providers. The Interim Measures for the Management of Generative Artificial Intelligence Services establish governance expectations for providers of generative AI services [2]. The Provisions on the Administration of Deep Synthesis Internet Information Services address management of deep synthesis services and synthetic media [3]. The Measures for the Labeling of Artificial Intelligence-Generated Synthetic Content and GB 45438-2025 define expectations for visible and metadata-based identification of AI-generated synthetic content [4], [5].

The labeling regime is especially relevant for machine-readable evidence. Industry commentary distinguishes between visible labeling and implicit or metadata-based labeling [14]. Visible labeling is user-facing and may appear as text, watermark, audio prompt, or interface notice. Implicit labeling is embedded in file metadata or comparable technical markers and is intended to support traceability and anti-tampering. The PwC report describes metadata elements such as generated-content label information, service-provider identifier, content identifier, dissemination-service provider identifier, and dissemination identifier. This supports the prototype's use of `visible_label_present`, `invisible_label_present`, and `metadata_label_present` as evidence fields rather than treating labeling as a single free-text disclosure.

Labeling obligations are also not limited to the point of generation. Industry commentary describes a responsibility chain across service providers, dissemination platforms, application distribution platforms, and users [14]. Dissemination platforms may need to verify metadata, process user declarations, identify synthetic traces, and provide declaration functions. This is why the prototype frames labeling as an operational evidence obligation connected to request records, output records, and labeling checks.

Other sources provide broader context. Algorithmic recommendation provisions address platform algorithm governance [15]. The Personal Information Protection Law, Data Security Law, and Cybersecurity Law define baseline expectations for personal information, data security, and network security governance [6], [16], [17]. For this prototype, these sources motivate a limited control set covering consent, scope, filing metadata, labeling, safety, and complaint handling.

III. CHINA AIGC COMPLIANCE EVIDENCE PROFILE

The prototype defines a compact evidence profile for machine-readable controls. Each control has a stable identifier, lifecycle phase, target type, evidence type, evaluation method, enforcement mode, threshold, responsible role, and remediation action. This profile is not a legal ontology. It is an engineering interface for connecting policy interpretation to operational logs and generated artifacts.

The profile supports repeatability. For example, a consent control can be evaluated by joining a generation request to an actor consent record and checking status, dates, usage scope, and region scope. A labeling control can be evaluated by joining an output to a labeling check. A complaint control can be evaluated by comparing complaint age or resolution time to a threshold.

IV. ARCHITECTURE: POLICY, EVIDENCE, AND ENFORCEMENT

The prototype uses a three-layer architecture, shown in Fig. 1. The layers are intentionally simple so that the relation between policy, operational evidence, and enforcement can be inspected.

Policy layer. The policy layer is a YAML control catalog. It defines control identifiers, descriptions, severity, and thresholds. In the current implementation, the policy file includes controls for consent, consent scope, model filing metadata, AIGC labeling, content safety, and complaint handling.

Evidence layer. The evidence layer contains synthetic operational logs and generated artifacts. Inputs include virtual actor records, actor consent records, model filing records, generation requests, output records, labeling checks, content safety checks, and complaint records. Outputs include assessment results, a risk register, remediation items, summary tables, and selected per-request evidence bundles.

Enforcement layer. The enforcement layer runs automated checks over the evidence layer. Each request receives a pass, review, or fail result. Review and fail conditions are aggregated into risk records and remediation items. This layer does not enforce legal compliance. It demonstrates how selected controls

TABLE I: Illustrative Mapping from Chinese AI Governance Concerns to Prototype Evidence Artifacts

Concern	Chinese regulatory source	Evidence obligation	Prototype artifact
Generative AI service governance	Interim Measures for the Management of Generative Artificial Intelligence Services	Demonstrate provider governance over generated content, service behavior, and safety processes.	Assessment results, content safety checks, risk register
Deep synthesis service governance	Provisions on the Administration of Deep Synthesis Internet Information Services	Maintain service governance for synthetic media and prevent unmanaged misuse of deep synthesis capabilities.	Generation request logs, output records, control catalog
AIGC visible and invisible labeling	Measures for the Labeling of Artificial Intelligence-Generated Synthetic Content; GB 45438-2025	Show that generated synthetic content carries visible and metadata-based labels where required.	Labeling check records, output-level evidence bundles
Model filing or registration disclosure	Generative AI service governance and filing practice	Record whether model filing or registration metadata exists for the model used in generation.	Model filing records, MODEL_FILING control result
Consent and asset authorization	Civil-law, personal information, and platform authorization expectations	Show that actor assets are covered by valid consent and that requested commercial use is within scope.	Actor consent records, consent scope controls
Personal information protection	Personal Information Protection Law of the People's Republic of China	Avoid processing real personal data in the prototype and preserve data minimization assumptions.	Synthetic-only metadata, dataset disclaimer
Content safety	Generative AI and platform content governance obligations	Route high-risk outputs to blocking or manual review rather than publication.	Content safety checks, assessment results, remediation items
Complaint handling and remediation	Platform accountability and complaint-handling obligations	Track complaints, status, SLA age, resolution evidence, and escalation items.	Complaint records, risk register, remediation items

TABLE II: Machine-Readable Evidence Profile

Property	Purpose	Example
control_id	Stable identifier for a machine-readable control.	AIGC_INVISIBLE_LABEL
regulation_source	Names the governance source that motivates the control.	Measures for AI-generated synthetic content labeling
article_ref	Records the article, clause, or standard section used for traceability.	Article or standard clause reference
lifecycle_phase	Places the control in the service lifecycle.	Pre-generation, publication, monitoring
target_type	Specifies the artifact being evaluated.	Consent record, model filing, output, complaint
evidence_type	Defines the evidence expected from the platform.	JSON log, label check, safety check, complaint record
evaluation_method	Describes how the control is evaluated.	Scope match, status check, threshold comparison
enforcement_mode	Indicates operational behavior when the control fails.	Monitor, review, block
risk_level	Captures severity for risk register construction.	Low, medium, high
threshold	Encodes the measurable rule.	Complaint resolved within 72 hours
threshold_type	Distinguishes binary, categorical, and numeric thresholds.	Boolean, enum, SLA duration
responsible_role	Identifies the operational owner.	Synthetic compliance operations
remediation_action	Links the failed control to an action item.	Suspend output until metadata label is present

Machine-Readable China AIGC Evidence Pipeline

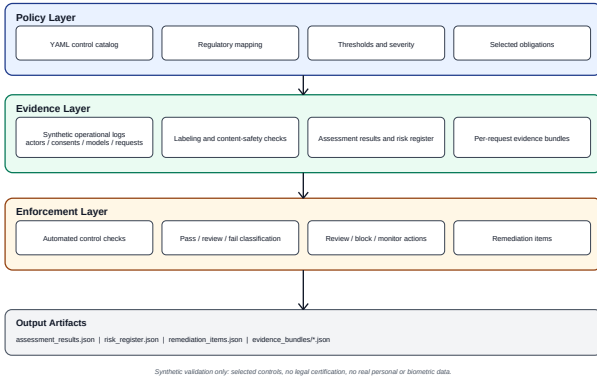


Fig. 1: Three-layer architecture for machine-readable AIGC compliance evidence.

can be evaluated consistently and converted into reviewable evidence.

V. VALIDATION SCENARIO: AIGC ACTOR AND DIGITAL HUMAN PLATFORM

The validation scenario models a platform that allows clients to generate advertising videos using licensed virtual actor assets. The scenario is designed around operational evidence rather than model performance. It asks whether a particular generation request can be supported by synthetic evidence of authorization, scope, labeling, content safety, and complaint handling.

The project does not use real faces, voices, contracts, company records, or personal information. All records are synthetic. During simulated operation, the platform checks actor consent, commercial scope, revoked consent, expired consent, model filing metadata, visible AIGC labels, invisible or metadata labels, content safety status, and complaint handling status.

VI. RESULTS

The seeded validation run used 1,000 synthetic generation requests. The dataset contained 50 synthetic virtual actor assets, 200 synthetic consent records, 10 synthetic model records, 1,000 output records, 1,000 labeling checks, 1,000

content safety checks, and 62 synthetic complaint records. The request-level assessment produced 800 pass outcomes, 120 review outcomes, and 80 fail outcomes. The artifact builder generated 88 selected per-request evidence bundles covering pass, review, fail, and complaint-related examples. The automated test suite contained 14 tests, all of which passed in the latest recorded run.

The distribution was calibrated for a synthetic validation scenario, not for a production benchmark. The pass rate was 80.0 percent, the review rate was 12.0 percent, and the fail rate was 8.0 percent. Failure conditions included expired consent, commercial use outside consent scope, revoked consent, missing model filing metadata, missing invisible labels, high-risk content that was allowed without blocking or manual review, and overdue complaint handling.

VII. DISCUSSION

The prototype highlights three points. First, China-facing AI compliance evidence obligations differ from EU AI Act high-risk conformity assessment. Chinese governance sources often emphasize generative AI service management, deep synthesis, labeling, model filing or registration disclosure, platform responsibility, complaint handling, and personal information protection.

Second, selected obligations can be decomposed into executable control logic. Consent existence becomes a status and validity check. Consent scope becomes a comparison between allowed and requested use. Labeling becomes an output-level evidence check. Complaint handling becomes a threshold-based status evaluation. These controls are modest, but they show how governance obligations can become operational evidence requirements.

Third, the value of the prototype is not that it provides a legal conclusion. It does not ensure compliance, certify a service, or replace counsel. Its value is to demonstrate how selected obligations can be transformed into a structured evidence pipeline that produces assessment results, risk records, remediation items, and per-request evidence bundles for technical review.

VIII. LIMITATIONS

This work has several limitations. The dataset is synthetic only. The prototype provides no legal certification and has no formal regulatory endorsement. It uses no real biometric, face, or voice data. It covers selected controls only and is not full Chinese AI compliance coverage. It is not sector-specific for finance, healthcare, education, media, or other regulated domains. It does not implement actual model training, production watermarking, cryptographic signing, or production-grade evidence retention. The regulatory mapping is illustrative and would require legal review before operational use.

IX. CONCLUSION

This paper presents a technical prototype for machine-readable compliance evidence in China-facing AIGC actor and digital human platforms. The prototype demonstrates feasibility for translating selected AI governance obligations into

controls, assessment results, risk records, remediation items, and per-request evidence bundles. Future work could expand the control catalog, integrate evidence integrity mechanisms, support more lifecycle phases, and connect the evidence profile to formal governance workflows.

APPENDIX A EXAMPLE CONTROL IN YAML

Listing 1: Example synthetic control rule.

```
controls:
  invisible_label:
    id: AIGC_INVISIBLE_LABEL
    description: Output has invisible AIGC provenance metadata.
    severity: medium
    lifecycle_phase: publication
    target_type: output
    evidence_type: labeling_check
    evaluation_method: metadata_label_present
    enforcement_mode: block_until_labeled
```

APPENDIX B EXAMPLE EVIDENCE BUNDLE SCHEMA

Listing 2: Simplified per-request evidence bundle schema.

```
{
  "metadata": {
    "bundle_type": "per_request_synthetic_evidence",
    "data_classification": "synthetic_operational_logs",
    "contains_real_personal_data": false
  },
  "assessment_result": {
    "request_id": "REQ-0001",
    "overall_status": "pass",
    "checks": []
  },
  "actor": {},
  "consent": {},
  "model": {},
  "generation_request": {},
  "output": {},
  "labeling_check": {},
  "content_safety_check": {},
  "complaint": null
}
```

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TABLE III: Control-Level Validation Results for the Synthetic AIGC Actor Platform

Phase	Control	Actual	Threshold	Result
Dataset generation	Generation requests assessed	1,000	1,000 synthetic requests	Met
Outcome calibration	Pass outcomes	800 (80.0%)	75–85%	Met
Outcome calibration	Review outcomes	120 (12.0%)	8–15%	Met
Outcome calibration	Fail outcomes	80 (8.0%)	5–10%	Met
Pre-generation	Consent existence exceptions	31	Captured as review/fail	Captured
Pre-generation	Consent scope exceptions	14	Captured as review/fail	Captured
Deployment	Model filing metadata exceptions	9	Captured as review/fail	Captured
Publication	Visible AIGC label exceptions	66	Captured as review/fail	Captured
Publication	Invisible / metadata label exceptions	11	Captured as review/fail	Captured
Publication	Content safety exceptions	67	Captured as review/fail	Captured
Monitoring	Complaint handling exceptions	15	Captured as review/fail	Captured
Evidence packaging	Selected per-request evidence bundles	88	Pass + review + fail + complaint-related requests	Met

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